

SHREYANSH SHUKLA

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Summary

I build end-to-end web applications, from pixel-precise frontends in React to robust backend in Python, high-speed low-latency REST APIs in Fastapi with a focus on performance, scalability, and clean architecture. I am proficient in developing both monolithic and microservice architecture, responsive web interfaces with hands-on experience with PostgreSQL, Docker and AWS.

Projects

Traffic Signal Optimization

Tech Stack: Python, FastApi, React, TailwindCSS, Supabase, Numpy, Matplotlib, PyTorch, Ultralytics

- Developed a **responsive web application with real time visual dashboard** using React/Tailwind. Leveraged **Supabase Realtime** to display live traffic signal updates and signal status. Allowing operators to monitor system performance at a glance.
- Engineered backend architecture using FastAPI to facilitate **high-speed communication** between the traffic analysis engine and the user interface, ensuring **low-latency data updates** for signal timing adjustments.
- Implemented a **robust data synchronization layer using Supabase** (PostgreSQL) to log traffic metrics and system events, enabling efficient storage, retrieval, and historical analysis of traffic patterns to support data-driven optimization.

LEXIENT (AI-Powered Second Brain System)

Tech Stack: React, TailwindCSS, FastApi, MongoDB

- Architected a modular and intuitive dashboard for an AI-powered knowledge management platform, using **React** to seamlessly integrate complex features like real-time reading trackers and interactive topic heatmaps.
- Designed a **context-aware UI** for AI-driven feedback, implementing responsive components that display LLM-generated flashcards and reflective prompts, ensuring an engaging, **low-latency** user experience.
- Engineered highly interactive data visualizations using Visualization Library **Recharts** to render timelines, transforming complex metadata into a visually coherent intellectual journey that enhances user insight.

Customer Segmentation by Purchasing Behavior

Tech Stack: Python, Pandas, Matplotlib, Seaborn, Scikit-learn

- Executed end-to-end customer segmentation by applying **K-Means clustering** to e-commerce transaction datasets, enabling targeted marketing strategies.
- Engineered a robust **RFM model** to analyze and quantify purchasing behavior, resulting in clear, actionable differentiation between customer groups.
- Improved the ability to identify high-value segments and at-risk customers.

Skills

Languages: Python, Java, JavaScript, SQL

Frontend: HTML, CSS, Tailwind CSS, React, Bootstrap, Axios

Backend: REST APIs, WebSockets, FastAPI, NodeJs

Database: MySQL, MongoDB, PostgreSQL

Cloud & DevOps: AWS, Docker, Git, YAML

Fundamentals: OOPS, DBMS, OS, DSA, System Design

Tools: VS Code, Postman, Linux/Unix

Certifications & Achievements

- Cloud Computing - NPTEL(2026)**
- Data Analyst - oneroadmap (2025)**
- Secured **3ed rank** in coding competition at **Code Mantra in Technovision 2025**
- Presented the research paper "Traffic Signal Optimization using Fuzzy logic" at NCACSIT-2025 National Conference.

Education

Samrat Ashok Technological Institute

Bachelor of Technology in Artificial Intelligence and Data Science

Nov. 2022 – May 2026

Vidisha, MP